Five classical theorems on triangles and simplices, formalised in Lean 4 with Mathlib: Carnot, Viviani, De Gua, the Japanese theorem and Poncelet's porism

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Abstract

We report Lean 4 formalisations, built on Mathlib, of five classical results of elementary geometry: Carnot's theorem $\sum_i d_i(O) = R + r$ for a triangle; Viviani's theorem, generalised to simplices of every positive dimension as the identity $\sum_i d_i(p)/h_i = 1$; De Gua's theorem in every dimension, for simplices whose edges at one vertex are pairwise orthogonal (with k-volume defined by a Gram determinant), together with the projection form obtained from a self-contained proof of the Cauchy–Binet formula; the Japanese theorem for convex cyclic polygons, for all triangulations by maximal sets of non-crossing diagonals; and Poncelet's porism for triangles, with its converse, in the "construction" form. The developments follow two strands. Viviani, Carnot and the Japanese theorem are statements about signed distances to the facets of a simplex, equivalently barycentric coordinates, and Poncelet's porism rests on the same incentre and excentre vocabulary together with circle tangency; here Mathlib's `Affine.Simplex` API (`signedInfDist`, `height`, `incenter`, `excenter`, `circumcenter`) supplies most of what is needed. De Gua's theorem belongs to a second strand, Gram determinants of edge vectors. We state every hypothesis exactly, explain why each is present, and give lemmas that tie the formal definitions to the familiar informal ones. Together the developments have about 3 200 lines of new Lean code (plus about 900 lines of verbatim copies; the Layer-14/15 check files added in version 2 have about 2 200 more), with no `sorry` and only Lean's standard axioms. As far as we could determine, the geometric signed-distance formulation of Carnot's theorem, the n-dimensional Viviani and De Gua theorems in this form, the Japanese theorem and Poncelet's porism have no earlier complete Lean formalisation; closely related Lean material does exist for the trigonometric identity underlying Carnot's theorem, for the three-dimensional De Gua theorem, for coordinate versions of Viviani's theorem and for the Cauchy–Binet formula, and we describe it in Section 9.

1 Introduction

The five theorems of this paper are classical, short, and familiar from olympiad geometry. They are all listed in Mathlib's "1000+ theorems" file [10], and at the Mathlib version we use none of them has a declaration recorded there. They are:

- *Carnot's theorem.* In a triangle with circumcentre O, circumradius R and inradius r, the signed distances from O to the three side lines sum to $R + r$.

- *Viviani's theorem.* In an equilateral triangle the distances from an interior point to the three sides sum to the altitude.
- *De Gua's theorem.* If the three edges of a tetrahedron at one vertex are pairwise perpendicular, the square of the area of the opposite face is the sum of the squares of the areas of the other three faces [2].
- *The Japanese theorem.* For a convex polygon inscribed in a circle, the sum of the inradii of the triangles of a triangulation does not depend on the triangulation [6].
- *Poncelet's porism for triangles.* Let Γ and ω be circles with radii $R > r > 0$ and centres at distance d. If $d^2 = R(R - 2r)$, then every point of Γ is a vertex of a triangle inscribed in Γ and circumscribed about ω. Conversely, if the centre of ω is inside Γ and one such triangle exists, then $d^2 = R(R - 2r)$ [5]. In this paper "circumscribed about ω" means that the three side *lines* are tangent to ω; see Section 7 for how this relates to the classical reading.

Two strands. Four of the five theorems use the vocabulary of *signed distances* from a point to the facets of a simplex; De Gua's theorem uses Gram determinants instead. For a simplex s with vertices $p_0, \ldots, p_n$, let $d_i(p)$ be the signed distance from p to the hyperplane of the facet opposite p_i, positive on the side of p_i, and let $h_i = d_i(p_i)$ be the i-th height. If $p = \sum_i w_i p_i$ with $\sum_i w_i = 1$, then

$$d_i(p) = w_i \, h_i. \tag{1}$$

Mathlib provides d_i as `Affine.Simplex.signedInfDist` and (1) as `Affine.Simplex.signedInfDist_affineCombination`. The roles of the two strands are as follows:

- *Viviani* is the statement $\sum_i d_i/h_i = \sum_i w_i = 1$, specialised to equal heights.
- *Carnot* is (1) for the circumcentre, whose barycentric weights are known, followed by a polynomial identity in the side lengths.
- *The Japanese theorem* is Carnot's theorem applied to each triangle of the triangulation. The contributions of each diagonal cancel, since a diagonal is traversed in opposite directions by its two triangles and the signed distance changes sign.
- *Poncelet's porism* for triangles does not follow from (1): the forward direction is an explicit coordinate construction of tangent lines to a circle. What it shares with the strand above is the incentre/excentre vocabulary. Its converse rests on Euler's formula $OI^2 = R(R - 2r)$, a statement about the incentre, the point with $d_0 = d_1 = d_2 = r$: the centre of ω is equidistant from the three side lines, so it is the incentre or an excentre.
- *De Gua's theorem* forms a second strand. Its natural language is Gram determinants of edge vectors rather than signed distances, and the general form is the Cauchy–Binet formula. It is a companion result, not another application of (1).

The actual proof reuse is Carnot → Japanese (the Japanese development imports a copy of the Carnot files), and Viviani and the Poncelet converse share Mathlib's simplex, incentre and excentre infrastructure.

Contributions.

- Lean formalisations of all five theorems, with the exact statements given in Sections 3–7 and indexed in Table 1.
- Generalisations where they come cheaply: Viviani for simplices of every positive dimension, with a characterisation of the simplices for which the distance sum is constant; De Gua in every dimension and in every real inner product space, with volume given by Definition 9; the Japanese theorem for *every* triangulation, defined geometrically as a maximal set of chords with pairwise disjoint interiors; Poncelet's porism in the form "for ω centred inside Γ, the construction from one point closes if and only if it closes from every point, if and only if $d^2 = R(R - 2r)$".
- A self-contained Lean proof of the Cauchy–Binet formula for rectangular matrices over a commutative ring, which is not in Mathlib at the version we use (Section 5).
- Correctness lemmas that connect each formal definition with its informal reading (Section 8).

Availability. The five developments are separate Lake projects, published at `https://github.com/jpbald93/carnot`, `https://github.com/jpbald93/viviani`, `https://github.com/jpbald93/degua`, `https://github.com/jpbald93/japanese-theorem` and `https://github.com/jpbald93/poncelet-triangles`. Line references in this paper are to commits `98e5614` (Carnot), `029aa3d` (Viviani), `d46e6b2` (De Gua), `34fd0a9` (Japanese) and `9affe62` (Poncelet). Each repository has a README with build and check commands. All five use Lean `v4.33.1` and the same Mathlib revision (`0df444a3`, tag `v4.33.1`), pinned in `lake-manifest.json`. The Japanese development contains a verbatim copy of the Carnot development under `Japanese/Carnot/` (only the import prefix differs). The Poncelet development contains, under `Poncelet3/Euler/`, a verbatim copy of three files from the author's separate formalisation of Feuerbach's theorem, which provides Euler's formula (again only the import line differs). Version 2 of this paper adds the Layer-14 and Layer-15 files of Section 10.1 in a new commit of each repository: `fa8c37951d290d26db66ff83c358b09b156d26e6` (Carnot), `e71460f2edb224b2e7e98a39c6db6bead93613cc` (Viviani), `daa89522f4a85e6afb8fe4d4d1a94b3d0761943e` (De Gua), `be4503b4e1dfdf7e527e2425276f992ecabb4bc2` (Japanese) and `aa9da6c4e077cee2e9191ad9011a246ad9e04f2f` (Poncelet). The earlier statements and library files are unchanged in that commit (only the root file gains two import lines, and `gate.sh` and the README are updated), so the line references above are valid at both commits. Supporting material (the Layer-14 and Layer-15 reports, the blind writer's statements and report, the transcript audit, and the reviews of this version with the fixes made in response) is in the folder `Triangle-Simplex` of the repository `https://github.com/jpbald93/LEAN-verification`.

Table 1: Main formal statements, grouped by repository. Links point to the declaration line at the commit of that repository listed under Availability.

Lean name	In this paper	Source
Repo `carnot`		
`Carnot.signedSideDist`	Def. 1	`Main.lean`:27
`Carnot.signedSideDist_eq_signedInfDist`	Lemma 29	`Main.lean`:49
`Carnot.signedSideDist_incenter`	Lemma 29	`Main.lean`:65

Table 1, continued

Lean name	In this paper	Source
`Carnot.carnot`	Theorem 2	`Main.lean`:75
`Carnot.carnot_sum`	Theorem 2	`Main.lean`:97
`Carnot.signedSideDist_circumcenter_eq`	Theorem 3	`Main.lean`:113
`Carnot.cos_add_cos_add_cos`	Theorem 3	`Main.lean`:151
`Carnot.carnot_poly`	Section 3	`Algebra.lean`:32
Repo ***viviani***		
`Affine.Simplex.sum_signedInfDist_div_height`	Theorem 4	`Basic.lean`:37
`Affine.Simplex.sum_signedInfDist_eq_height`	Theorem 5	`Heights.lean`:34
`Affine.Simplex.exists_forall_sum_signedInfDist_eq_iff`	Theorem 5	`Heights.lean`:43
`Affine.Simplex.signedInfDist_eq_dist_of_mem_closedInterior`	Lemma 30	`Heights.lean`:64
`Affine.Simplex.sum_dist_eq_height`	Theorem 5	`Heights.lean`:73
`Affine.Simplex.Equilateral.height_eq`	Proposition 6	`Equilateral.lean`:62
`Affine.Simplex.Regular.sum_dist_eq_height`	Corollary 7	`Equilateral.lean`:92
`Viviani.equilateral_triangle'`	Corollary 7	`Equilateral.lean`:109
`Affine.Simplex.Equilateral.height_eq` + `Affine.Simplex.sum_dist_eq_height`	Corollary 7 (equilateral case)	`Equilateral.lean`:62, `Heights.lean`:73
Repo ***degua***		
`DeGua.de_gua_three`	Theorem 8	`Three.lean`:42
`DeGua.simplexVolSq`	Def. 9	`General.lean`:97
`DeGua.det_diagonal_add_const`	Lemma 11	`General.lean`:45
`DeGua.de_gua`	Theorem 10	`General.lean`:130
`DeGua.triAreaSq_eq_simplexVolSq`	Lemma 31	`Bridge.lean`:16
`DeGua.de_gua_three_of_general`	Lemma 31	`Bridge.lean`:28
`DeGua.det_mul_eq_sum_strictMono`	Theorem 12	`CauchyBinet.lean`:33
`DeGua.det_gram_eq_sum_sq_minors`	Corollary 13	`CauchyBinet.lean`:91
Repo ***japanese***		
`Japanese.chordDist`	Def. 14	`Basic.lean`:30
`Japanese.chordDist_swap`	Lemma 15	`Basic.lean`:35
`Japanese.triInradius_eq`	Lemma 16	`Basic.lean`:130
`Japanese.japanese_quadrilateral`	Theorem 17	`Basic.lean`:193
`Japanese.Triangulation`	Def. 18	`Polygon.lean`:33
`Japanese.inradiusSum_eq`	Theorem 19	`Polygon.lean`:97
`Japanese.japanese_polygon`	Theorem 19	`Polygon.lean`:131
`Japanese.IsTriangulation`	Def. 20	`Combinatorial.lean`:30
`Japanese.isTriangulation_iff_exists_tree`	Proposition 21	`Combinatorial.lean`:415
`Japanese.Triangulation.chords_injective`	Proposition 21	`Combinatorial.lean`:453
`Japanese.treeEquiv`	Proposition 21	`Combinatorial.lean`:481
`Japanese.toFinset_triangles`	Proposition 21 (triangles = `triangleSet`)	`Combinatorial.lean`:259
`Japanese.card_triangleSet`	Proposition 21 ($j - i - 1$ triangles)	`Combinatorial.lean`:265
`Japanese.japanese_polygon_combinatorial`	Theorem 23	`Combinatorial.lean`:520
`Japanese.sbtw_inter_iff_interleave`	Proposition 22	`Geometric.lean`:182
`Japanese.geomCrosses_iff`	Proposition 22	`Geometric.lean`:220
`Japanese.IsGeomTriangulation`	Def. 20	`Geometric.lean`:209
`Japanese.japanese_polygon_geometric`	Theorem 23	`Geometric.lean`:276
Repo ***poncelet3***		

Table 1, continued

Lean name	In this paper	Source
`Feuerbach.dist_circumcenter_incenter_sq`	Theorem 24	`Main.lean`:100
`Feuerbach.dist_circumcenter_excenter_sq`	Theorem 24	`Main.lean`:105
`Poncelet3.exists_poncelet_triangle`	Theorem 25	`Porism.lean`:104
`Poncelet3.euler_of_isTangent`	Theorem 26	`Closure.lean`:35
`Poncelet3.poncelet_triangle_iff`	Theorem 26	`Closure.lean`:105
`Poncelet3.radius_lt_dist_center_of_mem`	Lemma 32	`Construction.lean`:77
`Poncelet3.tangentsFrom_eq_pair`	Lemma 32	`Construction.lean`:263
`Poncelet3.secondInter_spec`	Lemma 32	`Construction.lean`:313
`Poncelet3.poncelet_construction`	Theorem 28	`Construction.lean`:334
`Poncelet3.ConstructionCloses`	Def. 27	`Package.lean`:37
`Poncelet3.poncelet_construction_iff`	Theorem 28	`Package.lean`:98
`Poncelet3.poncelet_all_or_nothing`	Theorem 28	`Package.lean`:107

2 Setting and Mathlib vocabulary

Throughout, V is a real inner product space and P a metric affine space over it:

```
[NormedAddCommGroup V] [InnerProductSpace ℝ V] [MetricSpace P] [NormedAddTorsor V P]
```

No finite-dimensionality is assumed unless stated. The Japanese theorem and Poncelet's porism additionally assume `[Fact (finrank ℝ V = 2)]`, so that P is a Euclidean plane; the Japanese theorem also fixes an orientation `o : Orientation ℝ V (Fin 2)`, with area form ω_o (`Orientation.areaForm`).

We use the following Mathlib notions for a simplex `s : Affine.Simplex ℝ P n`; an `Affine.Triangle` is the case $n = 2$. Heights and signed facet distances require $n \geq 1$ (Mathlib's `[NeZero n]`), since a 0-simplex has no facets.

- `Affine.Simplex.height s i` is the distance from the i-th vertex to the affine span of the opposite facet; `Affine.Simplex.height_pos` says it is positive.
- `Affine.Simplex.signedInfDist s i p` is the signed distance from p to the affine span of the facet opposite vertex i, positive towards vertex i. Components of p orthogonal to the span of the simplex are ignored.
- `Affine.Simplex.circumcenter`, `Affine.Simplex.circumradius`, `Affine.Simplex.incenter`, `Affine.Simplex.inradius`, `Affine.Simplex.excenter` and `Affine.Simplex.exradius` have their usual meaning. The incentre and excentres are defined for general simplices through weights (`Affine.Simplex.excenterWeights`).
- `Affine.Simplex.closedInterior` is the set of affine combinations with weights in $[0, 1]$, that is, the closed simplex.
- `Affine.Simplex.Equilateral` means all edges have the same length, and `Affine.Simplex.Regular` means every permutation of the vertices is induced by an isometry.

Circles are `EuclideanGeometry.Sphere P`. An affine subspace is tangent to a sphere (`EuclideanGeometry.Sphere.IsTangent`) if it lies in the tangent space (`EuclideanGeometry.Sphere.orthRadius`) at some point of the sphere that it contains. `EuclideanGeometry.Sphere.tangentsFrom A` is the set of maximal tangent spaces through A, and `EuclideanGeometry.`

`Sphere.secondInter` is the second intersection of a line with a sphere. We write `line[ℝ, A, B]` for the affine span of $\{A, B\}$.

3 Carnot's theorem

Definition 1. For a triangle t and $i \in \{0, 1, 2\}$, let F_i be the foot of the altitude from vertex A_i and h_i the height. The signed side distance of p is

$$\texttt{signedSideDist}\ t\ i\ p = \langle p - F_i,\ A_i - F_i \rangle / h_i,$$

the component of $p - F_i$ along the unit normal to side i that points towards A_i.

It agrees with Mathlib's `signedInfDist` on the plane of the triangle (Lemma 29), which is where the circumcentre lies. The key property is `Carnot.signedSideDist_affineCombination`, the triangle case of (1).

Theorem 2 (Carnot)**.** *For every triangle t in P,*

$$\sum_{i=0}^{2} \texttt{signedSideDist}\ t\ i\ O = R + r,$$

where O, R, r are the circumcentre, circumradius and inradius of t.

```
theorem carnot :
    signedSideDist t 0 t.circumcenter + signedSideDist t 1 t.circumcenter +
        signedSideDist t 2 t.circumcenter = t.circumradius + t.inradius
```

There are no hypotheses beyond t being a triangle, which in Mathlib includes affine independence. There is no dimension hypothesis. The distances are signed, and this is essential: for an obtuse triangle the circumcentre lies outside, and the distance to the side opposite the obtuse angle is negative. Theorem 3 makes the sign convention visible.

Theorem 3. *Let α_i be the angle of t at vertex A_i. Then* `signedSideDist` $t\ i\ O = R\cos\alpha_i$ *for each i, and $\cos\alpha_0 + \cos\alpha_1 + \cos\alpha_2 = 1 + r/R$.*

Proof. The circumcentre is written as $O = \sum_i w_i A_i$. With side lengths a, b, c and $k = 2\cdot$area, one has $h_i = k/a_i$, $2kR = abc$ and, from `Affine.Simplex.inradius_eq_abs_inv_sum`, $r = k/(a+b+c)$. By (1), Theorem 2 becomes $\sum_i w_i k/a_i = R + k/(a+b+c)$, which, after clearing denominators, is the polynomial identity `Carnot.carnot_poly`:

$$(w_0 bc + w_1 ca + w_2 ab)(a+b+c) = 2R^2(a+b+c) + abc.$$

It holds modulo the three equations $|O - A_i|^2 = R^2$, written in the side lengths and the weights. The coefficients of the `linear_combination` certificate (of degree two) were found with a computer algebra system. The cosine form uses three further certificates of degree three and the law of cosines from Mathlib. The frame machinery (`Carnot.frame_data`, `Carnot.height_sq_mul_sq`) is adapted from the author's Feuerbach development.

4 Viviani's theorem, generalised to simplices

Here s is an n-simplex with $n \geq 1$ (`[NeZero n]`), in any real inner product space. We write $d_i(p)$ for `s.signedInfDist i p` and h_i for `s.height i`.

Theorem 4. *For every point p in the affine span of s, $\sum_i d_i(p)/h_i = 1$. Moreover, $d_i(p)/h_i$ is the i-th barycentric coordinate of p.*

Proof. Write $p = \sum_i w_i p_i$ with $\sum_i w_i = 1$, and apply (1). □

Theorem 5. *Suppose all heights of s are equal to h. Then $\sum_i d_i(p) = h$ for every p in the affine span of s, and $\sum_i \operatorname{dist}(p, \Pi_i) = h$ for every p in the closed simplex, where Π_i is the affine span of the i-th facet. Conversely, if $p \mapsto \sum_i d_i(p)$ is constant on the affine span of s, then all heights are equal.*

```
theorem sum_signedInfDist_eq_height (hh : ∀ i j, s.height i = s.height j)
    {p : P} (hp : p ∈ affineSpan ℝ (Set.range s.points)) (k : Fin (n + 1)) :
    Σ i, s.signedInfDist i p = s.height k
theorem exists_forall_sum_signedInfDist_eq_iff :
    (∃ c : ℝ, ∀ p ∈ affineSpan ℝ (Set.range s.points), Σ i, s.signedInfDist i p = c) ↔
      ∀ i j, s.height i = s.height j
```

The converse is proved by evaluating the sum at two vertices: $\sum_i d_i(p_j) = h_j$.

Why each hypothesis. The signed form holds on the whole affine span, which is the natural domain of d_i. The unsigned form needs p in the closed simplex, where all $d_i(p) \geq 0$ (weights in $[0, 1]$); outside it, some distance must be counted negatively. We use Mathlib's `Affine.Simplex.closedInterior`, so the unsigned form includes boundary points. "Distance to a side" means the distance to the affine span of the facet (the side *line* of a triangle), attained at the orthogonal projection (`Affine.Simplex.orthogonalProjectionSpan`). This is the classical reading. For an equilateral triangle the foot of the perpendicular from a point of the closed triangle lies on the side itself, but for a general simplex it need not, and we do not use or prove any such fact.

Proposition 6. *All heights of an equilateral simplex are equal.*

Proof. For $i \neq j$, reflect in the perpendicular bisector hyperplane of $[p_i, p_j]$. This affine isometry (`Viviani.perpReflection`) swaps p_i and p_j and fixes every other vertex, since those are equidistant from p_i and p_j. Heights are invariant under isometries (`Affine.Simplex.height_map`) and permute under reindexing (`Affine.Simplex.height_reindex`), so $h_i = h_j$. □

Corollary 7 (Viviani)**.** *For a regular (or just equilateral) simplex, and every point p of the closed simplex, the distances from p to the facet hyperplanes sum to the height. For an equilateral triangle `t` and `p ∈ t.closedInterior`, the three distances sum to `t.height 0`.*

Equal heights do not imply equilateral when $n \geq 3$. Since $h_i \cdot \operatorname{vol}(F_i) = n \operatorname{vol}(s)$, equal heights means equal facet volumes. In dimension three this holds for every tetrahedron with four congruent faces (a disphenoid), which need not be regular. So Theorem 5 characterises the simplices with constant distance sum: they are the simplices with equal heights, a strictly larger class than the equilateral ones. The identity $h_i \operatorname{vol}(F_i) = n \operatorname{vol}(s)$ is not formalised here.

5 De Gua's theorem and the Cauchy–Binet formula

5.1 The three-dimensional statement

In $\mathbb{R}^3$ (`EuclideanSpace ℝ (Fin 3)`), let `DeGua.triAreaSq` $X\,Y\,Z = |(Y-X)\times(Z-X)|^2/4$, the squared area of a triangle.

Theorem 8. *If $A-O$, $B-O$, $C-O$ are pairwise orthogonal, then* `triAreaSq` $A\,B\,C =$ `triAreaSq` $O\,A\,B +$ `triAreaSq` $O\,B\,C +$ `triAreaSq` $O\,C\,A$.

The proof is one `linear_combination` (`DeGua.cross_sub_sq`).

5.2 All dimensions

There is no cross product in general, so we measure volume by Gram determinants.

Definition 9. For points $x_0,\dots,x_k$ of a real inner product space E,

$$\texttt{simplexVolSq}\ x = \det\big(\langle x_i - x_0,\, x_j - x_0\rangle\big)_{1\le i,j\le k}/(k!)^2.$$

This is the standard squared k-volume of the simplex $x_0\cdots x_k$. We stress that it is a *definition*: we have not related it to a measure. Mathlib now has a Euclidean Hausdorff measure (`MeasureTheory.Measure.eucliedeanHausdorffMeasure`), and connecting the two would need the facts that a parallelotope has volume $\sqrt{\det G}$ and a simplex $1/k!$ of that.

Theorem 10 (De Gua, general form)**.** *Let $O \in E$ and $P_0,\dots,P_m \in E$ with $P_i - O$ pairwise orthogonal. Then*

$$\texttt{simplexVolSq}\,(P_0,\dots,P_m) = \sum_{i=0}^{m} \texttt{simplexVolSq}\,(O,P_0,\dots,\widehat{P_i},\dots,P_m).$$

```
theorem de_gua (O : E) {m : ℕ} (P : Fin (m + 1) → E)
    (h : Pairwise fun i j => inner ℝ (P i - O) (P j - O) = 0) :
    simplexVolSq P =
      Σ i : Fin (m + 1), simplexVolSq (Fin.cons O (P ∘ i.succAbove) : Fin (m + 1) → E)
```

The only hypothesis is orthogonality. The points need not be distinct and E need not be finite-dimensional, and degenerate configurations (for instance a zero edge) are allowed.

Lemma 11. *For a finite index type ι, $b \in \mathbb{R}^\iota$ and $c \in \mathbb{R}$, with J the all-ones matrix,* $\det(\mathrm{diag}(b) + cJ) = \prod_i b_i + c\sum_i \prod_{j\neq i} b_j$.

When all $b_i \neq 0$, this follows from Mathlib's matrix determinant lemma `Matrix.det_one_add_replicateCol_mul_replicateRow`. For general b, replace b by $b + t\mathbf{1}$. Both sides are continuous in t, and they agree whenever $t \neq -b_i$ for all i, that is, on the complement of a finite set. That complement is dense, so they agree at $t = 0$. The Gram matrix of the edges $P_j - P_0$ of the facet opposite O is $\mathrm{diag}(|P_j - O|^2)_{j\ge 1} + |P_0 - O|^2 J$, so Lemma 11 gives $\sum_i \prod_{k\neq i} |P_k - O|^2$. The facets through O have diagonal Gram matrices with determinants $\prod_{k\neq i} |P_k - O|^2$.

5.3 Cauchy–Binet and the projection form

Theorem 12 (Cauchy–Binet)**.** *Let R be a commutative ring, A a $k \times n$ and B an $n \times k$ matrix over R. Then*

$$\det(AB) = \sum_S \det(A_{\cdot,S}) \det(B_{S,\cdot}),$$

the sum over strictly increasing $S : \{0..k-1\} \to \{0..n-1\}$, i.e. over k-subsets of columns.

```
theorem det_mul_eq_sum_strictMono (A : Matrix (Fin k) (Fin n) R)
    (B : Matrix (Fin n) (Fin k) R) :
    det (A * B) = Σ s ∈ univ.filter (fun s : Fin k → Fin n => StrictMono s),
      det (A.submatrix id s) * det (B.submatrix s id)
```

The proof expands $\det(AB)$ as a sum over all maps $p : \operatorname{Fin} k \to \operatorname{Fin} n$, drops the non-injective ones (their minor has two equal columns), and re-indexes the injective ones as pairs (strictly monotone s, permutation τ) with Mathlib's `Tuple.sort` and `Tuple.unique_monotone`. It covers $k > n$ (both sides vanish) and $k = 0$.

Corollary 13 (Pythagorean theorem for k-volumes)**.** *For $v_0, \dots, v_{k-1} \in \mathbb{R}^n$, the Gram determinant of v equals the sum, over all coordinate k-planes, of the squared $k \times k$ minors of the matrix with rows v_i. That is, the squared volume of the parallelotope spanned by v is the sum of the squared volumes of its orthogonal projections onto the coordinate k-planes.*

For $k = n-1$, applied to the facet opposite the corner of a simplex whose corner edges lie along the coordinate axes, this recovers De Gua's theorem; this is Conant and Beyer's generalisation [2]. We have not formalised that deduction: Theorem 10 is proved directly. Theorem 12 is, as far as we could determine, absent from Mathlib at our pinned version, and a Mathlib pull request for it is open (Section 9). Ours may be useful as a second, short proof (∼70 lines); we make no stronger claim.

6 The Japanese theorem

Here P is an oriented Euclidean plane.

Definition 14. For points O, p, q, let `chordDist` O p q $= \omega_o(q - p, O - p)/|pq|$, for $p \neq q$ the signed distance from O to the directed line pq, positive if O is to the left of the chord $p \to q$ (the total Lean definition returns 0 when $p = q$). Let `Japanese.triInradius` $a\,b\,c$ be the inradius of the triangle abc, or 0 if a, b, c are collinear.

Lemma 15 (cancellation)**.** `chordDist` O q $p = -$`chordDist` O p q.

Lemma 16 (Carnot, chord form)**.** *If a, b, c lie on a circle with centre O and radius R and $\omega_o(b - a, c - a) > 0$, then* `triInradius` a b $c = d(ab) + d(bc) + d(ca) - R$, *where $d(pq) =$* `chordDist` O p q.

This is Theorem 2 (from the copied Carnot files), after identifying the circle with the circumcircle (the triangle spans the plane) and Carnot's inward signed distance with the chord distance along the counterclockwise side (`Japanese.signedSideDist_eq_chordDist`).

Theorem 17 (quadrilateral)**.** *If A, B, C, D lie on a circle and the triangles ABC, ACD, ABD, BCD are positively oriented, then $r(ABC) + r(ACD) = r(ABD) + r(BCD)$.*

Polygons and triangulations. A polygon is a map $p : \mathbb{N} \to P$ of which only $p_0, \dots, p_{n-1}$ are used. The hypotheses are

- `hs` : $\forall$ `l < n, p l` $\in$ `s` for a circle `s`, and
- `hconv`: $\omega_o(p_b - p_a,\ p_c - p_a) > 0$ whenever $a < b < c < n$.

For $n \geq 3$ the second says that the polygon is strictly convex and listed counterclockwise. For $n = 2$ it is vacuous (even $p_0 = p_1$ is allowed); the formal theorems handle this harmless degenerate case through empty sums of triangles. The hypothesis is needed because Lemma 16 requires each triangle to be positively oriented. For a negatively oriented triangle the three chord distances change sign, so the chord sum is $-(R+r)$ and the formula becomes $r = -d(ab) - d(bc) - d(ca) - R$; reversing the ambient orientation restores the stated form. The counterclockwise choice is thus a convention needed by the displayed signed formula, not a geometric restriction: the inradius sum itself does not depend on the orientation, and the clockwise case follows by reversing o. For points on a circle, `hconv` is the same as being in counterclockwise cyclic order, but we did not formalise this equivalence.

Definition 18. `Japanese.Triangulation` i j is the inductive type with constructors `edge i : Triangulation i (i+1)` and `node (i<k) (k<j) : Triangulation i k` $\to$ `Triangulation k j` $\to$ `Triangulation i j`. Its triangles (`Japanese.Triangulation.triangles`) are the triples (i, k, j) at the nodes.

A `node` is the triangle (i, k, j) standing on the side ij, together with triangulations of the two remaining sub-polygons. Summing Lemma 16 over the triangles, the diagonals ik and kj cancel by Lemma 15. Induction on the tree gives:

Theorem 19. *Under* `hs` *and* `hconv`*, every* T : `Triangulation` 0 $(n-1)$ *has inradius sum* $\sum_l d(p_l p_{l+1}) - (n-2)R$*, where the sum runs over the* n *sides of the polygon. In particular any two such* T, T' *have the same inradius sum.*

No flip-connectivity argument is needed: the value is computed for every tree directly.

Triangulations as sets of chords. The tree type is a convenient model, but a triangulation is usually defined as a maximal set of non-crossing diagonals. We prove that the two agree. A chord of $i..j$ is a pair (a, b) with $i \leq a < b \leq j$; the chords (a, b) and (c, d) *cross* if $a < c < b < d$ or $c < a < d < b$ (`Japanese.Crosses`).

Definition 20. E is a *triangulation* of $i..j$ (`Japanese.IsTriangulation`) if all elements are chords, no two cross, and every chord that crosses no element of E is in E. It is a *geometric triangulation* (`Japanese.IsGeomTriangulation`) if the same holds with "cross" replaced by "the open segments $p_a p_b$ and $p_c p_d$ meet" for $e \neq f$. The triangles of E (`Japanese.triangleSet`) are the triples (a, b, c) with ab, bc, $ac \in E$.

Sides of the polygon and the root side cross nothing, so by maximality they belong to E. Hence E minus the sides is exactly a maximal non-crossing set of diagonals, and triangles can be read off uniformly. Deriving the triangles from E, instead of axiomatising them, avoids conditions such as "each diagonal lies in exactly two triangles".

Proposition 21. *For* $i < j$*, the map* $T \mapsto$ `Japanese.Triangulation.chords` T *is a bijection from trees to triangulations of* $i..j$ *(*`Japanese.treeEquiv`*). Under it, the tree's triangles are exactly* `triangleSet`*, and every triangulation of* $i..j$ *has* $j - i - 1$ *triangles.*

The hard direction builds a tree from E. If $i+1<j$, let k be the largest $b<j$ with $(i,b)\in E$. Then (k,j) crosses no element of E, so it is in E, and every chord of E other than the root side (i,j) lies on one side of k (`Japanese.IsTriangulation.exists_apex`). Restricting E to both sides gives triangulations of $i..k$ and $k..j$. Strong induction on $j-i$ finishes the proof.

Proposition 22. *Under* `hconv`*, for distinct chords e, f of $0..n-1$, the open segments of e and f meet if and only if e and f cross. Consequently, geometric and combinatorial triangulations coincide.*

For interleaved indices the intersection point is written down explicitly as a ratio of area forms; for non-interleaved disjoint pairs both endpoints of one chord lie strictly on one side of the other; chords with a shared endpoint never meet in their interiors because three vertices are never collinear. The case $e=f$ is excluded because a segment meets itself.

Theorem 23 (Japanese theorem)**.** *Let $n\geq 2$, and assume* `hs` *and* `hconv`*. For any two geometric triangulations E, E' of the polygon $p_0\cdots p_{n-1}$, $\sum_{t\in\texttt{triangleSet}\,E} r(t)=\sum_{t\in\texttt{triangleSet}\,E'} r(t)$. The same holds for combinatorial triangulations.*

```
theorem japanese_polygon_geometric (s : Sphere P) (hn : 2 ≤ n)
    (hs : ∀ l < n, p l ∈ s)
    (hconv : ∀ a b c, a < b → b < c → c < n →
      0 < o.areaForm (p b -ᵥ p a) (p c -ᵥ p a))
    {E E' : Finset (ℕ × ℕ)} (hE : IsGeomTriangulation p 0 (n - 1) E)
    (hE' : IsGeomTriangulation p 0 (n - 1) E') :
    triangleInradiusSum p (triangleSet E) = triangleInradiusSum p (triangleSet E')
```

The hypothesis $n\geq 2$ only excludes degenerate cases. What is still not formalised is a dissection in the sense of "a finite family of triangles with disjoint interiors whose union is the polygon"; our geometric triangulations are maximal sets of non-crossing chords (Section 11).

7 Poncelet's porism for triangles

Here P is a Euclidean plane, Γ and ω are spheres with centres O, I and radii R, r, and $d=|OI|$.

Theorem 24 (Euler)**.** *For every triangle, $|OI|^2=R(R-2r)$ and $|OI_i|^2=R(R+2r_i)$, where I_i and r_i are the excentre and exradius opposite vertex i.*

The two formulas are `Feuerbach.dist_circumcenter_incenter_sq` and `Feuerbach.dist_circumcenter_excenter_sq`, copied from the author's Feuerbach development. Both are also proved in Wang's Feuerbach repository and open Mathlib pull request (Section 9).

Theorem 25 (porism)**.** *Assume $0<r$, $r<R$ and $d^2=R(R-2r)$. For every $A\in\Gamma$ there are $B,C\in\Gamma$, with A,B,C pairwise distinct, such that the lines AB, AC and BC are tangent to ω.*

This, and every Poncelet statement below, asserts tangency of the three side *lines*. It does not assert that ω is the incircle of ABC, that the points of tangency lie on the side segments, or that the disk bounded by ω lies in the triangle. *Informally*, that classical reading follows under the hypotheses of Theorem 25: they give $d<R-r<R$ (`Poncelet3.dist_center_lt`), so I is inside Γ; by the classification of points equidistant from the side lines and Euler's excentral formula used in the proof of Theorem 26, I cannot be an excentre, so ω is the incircle. In the main files this argument is not a formal output: the converse proof contains most of it, but the theorem it returns is

only Euler's relation. Version 2 proves it in the Layer-15 file, for I inside Γ: three tangent side lines make ω Mathlib's insphere of the triangle (`Poncelet3.Classical.insphere_eq_of_isTangent`, Section 10.1).

```
theorem exists_poncelet_triangle [Fact (Module.finrank ℝ V = 2)] (Γ ω : Sphere P)
    (hr : 0 < ω.radius) (hrR : ω.radius < Γ.radius)
    (hd : dist Γ.center ω.center ^ 2 = Γ.radius * (Γ.radius - 2 * ω.radius))
    {A : P} (hA : A ∈ Γ) :
    ∃ B C : P, B ∈ Γ ∧ C ∈ Γ ∧ A ≠ B ∧ A ≠ C ∧ B ≠ C ∧
      ω.IsTangent line[ℝ, A, B] ∧ ω.IsTangent line[ℝ, A, C] ∧ ω.IsTangent line[ℝ, B, C]
```

Why each hypothesis. Mathlib's spheres may have any real radius. Given $0 < r$ and $d^2 = R(R-2r)$, the hypothesis $r < R$ is equivalent to $R > 0$: if $R > 0$ then $R - 2r = d^2/R \geq 0$, so $R > r$. It therefore only excludes spheres of nonpositive radius, for which $R(R-2r) > 0$ can hold too. The hypothesis $r > 0$ excludes a point-circle. We do *not* assume that ω lies inside Γ, or that A lies outside ω: both follow (`Poncelet3.radius_lt_dist_center_of_mem` shows $r < |AI|$).

Proof. We use coordinates and polynomial certificates, not the classical synthetic argument. Take I as origin, $u = A - I$ and $v = Ju$ with J the rotation by a right angle, and set $p = |u|^2$. The points B, C and the three points of tangency are given by explicit formulas in u, v, r, R, $s = \sqrt{p - r^2}$ and the coordinates X, Y of $O - I$. The facts "on Γ", "on ω", "perpendicular to the radius", "collinear" and two non-degeneracy identities are 16 `linear_combination` identities modulo five relations, with coefficients found by Gröbner-basis reduction in a computer algebra system. Tangency then follows from a dimension count: in the plane, a line through two distinct points of ω's tangent space at T equals that tangent space.

Theorem 26 (converse)**.** *Suppose $A, B, C \in \Gamma$ are pairwise distinct, the lines AB, AC, BC are tangent to ω, and $|IO| < R$. Then $d^2 = R(R-2r)$. Hence, if $0 < r < R$ and $|IO| < R$, every point of Γ is a vertex of such a triangle if and only if $d^2 = R(R-2r)$.*

Why I must be inside Γ. The triangle ABC spans the plane, so Γ is its circumcircle, and I is at distance r from all three side lines. By Mathlib's `Affine.Simplex.exists_forall_dist_eq_iff_exists_excenterExists_and_eq_excenter`, I is the incentre or an excentre. In the incentre case, Euler's formula applies. In an excentre case, $d^2 = R(R + 2r_i) \geq R^2$, contradicting $|IO| < R$. Without this hypothesis the first statement of Theorem 26 is false. For example, for $A = (0, 0)$, $B = (1, \frac{1}{10})$, $C = (1, -\frac{1}{10})$ and ω the excircle opposite A, we have $r_A \approx 0.110 < R \approx 0.505$, all three side lines are tangent to ω, and $d^2 = R(R + 2r_A) \neq R(R - 2r_A)$.

Informal remark (not formalised). In the "every point" form, and only there, the inside-centre hypothesis can be dropped while keeping $0 < r < R$. By the argument above, any one triangle forces $d^2 = R(R-2r)$ or the excentral alternative $d^2 = R(R+2r)$. In the latter, $d^2 > R^2$, so $d > R$, and $(R+r)^2 - d^2 = r^2 > 0$, so $R < d < R + r$. The point of Γ on the ray from O towards I is at distance $d - R < r$ from I, hence strictly inside ω, and no tangent to ω passes through it. So the universal-vertex property rules out the excentral alternative. This does *not* remove the inside-centre hypothesis from the one-triangle converse, nor from the all-or-nothing equivalence of Theorem 28; the excircle example above shows that it cannot be removed from the former.

The construction form. The classical statement is about a construction: draw the two tangents from A to ω, let them meet Γ again at B and C; then BC is tangent to ω. Theorem 25 only asserts that *some* such B, C exist. We therefore also prove the construction form.

Definition 27. `Poncelet3.ConstructionCloses` $\Gamma\ \omega\ A$ holds if (i) there are two distinct lines $\ell_1, \ell_2 \in \omega$.`tangentsFrom` A and points $B \in \Gamma \cap \ell_1$, $C \in \Gamma \cap \ell_2$ different from A; and (ii) for *all* such $\ell_1 \neq \ell_2$, B, C, the line BC is tangent to ω.

Part (i) prevents the predicate from holding vacuously, for instance when A is inside ω or when a tangent from A touches Γ only at A.

Theorem 28. *Assume $0 < r < R$ and $|IO| < R$, and let $A \in \Gamma$. Then* `ConstructionCloses` $\Gamma\ \omega\ A$ *holds if and only if $d^2 = R(R-2r)$, and if and only if* `ConstructionCloses` $\Gamma\ \omega\ A'$ *holds for every $A' \in \Gamma$. Moreover, if $0 < r < R$ and $d^2 = R(R-2r)$, then for any two distinct tangents ℓ_1, ℓ_2 from A and any second intersections B, C, the line BC is tangent (*`Poncelet3.poncelet_construction`*).*

The one new algebraic step is completeness of the tangency points: if $T \in \omega$ and A lies on the tangent at T, then T is one of the two explicit tangency points. From it, `tangentsFrom` A has exactly two elements, and each meets Γ in exactly one point other than A.

8 Do the statements mean what they say?

A reader of the statements above must trust several definitions. We list the lemmas that support the intended reading.

Lemma 29 (Carnot)**.** *For p in the plane of t,* `signedSideDist` $t\ i\ p =$ `signedInfDist` $t\ i\ p$*. The vertex A_i is at signed distance $h_i > 0$ from side i (*`Carnot.signedSideDist_point_self`*), and the incentre is at signed distance r from every side (*`Carnot.signedSideDist_incenter`*).*

Together with Theorem 3 ($d_i(O) = R\cos\alpha_i$) this pins down the sign convention: the distance is positive on the side of the interior, and negative for the circumcentre exactly when the opposite angle is obtuse. The circumcentre lies in the plane of t by Mathlib's `Affine.Simplex.circumcenter_mem_affineSpan`, so Carnot's theorem could equally be stated with Mathlib's `signedInfDist`; the main files do not, but the Layer-14 file does (`Carnot.l14_carnot`, Section 10.1).

Lemma 30 (Viviani)**.** *For p in the closed simplex, $0 \le d_i(p)$ and $d_i(p) = \operatorname{dist}(p,\ \pi_i(p))$, where π_i is the orthogonal projection onto the affine span of the i-th facet.*

So the unsigned Viviani statement uses ordinary distances, computed from Mathlib's `Affine.Simplex.faceOpposite` and `Affine.Simplex.orthogonalProjectionSpan`, and does not depend on our choice of signed distance.

Lemma 31 (De Gua)**.** *In $\mathbb{R}^3$,* `triAreaSq` $X\ Y\ Z =$ `simplexVolSq` (X, Y, Z) *(Lagrange's identity), and Theorem 8 is the case $m = 2$ of Theorem 10 (*`DeGua.de_gua_three_of_general`*).*

So the Gram-determinant volume agrees with the cross-product area in the one case where both are familiar. A reader who is not convinced by Definition 9 can rely on Theorem 8, whose definition of area is elementary.

Japanese theorem. Proposition 21 and Proposition 22 are correctness lemmas: they show that the tree model, the combinatorial definition and the geometric definition of a triangulation all describe the same objects, and that the triangles are the expected ones, $n-2$ in number (`Japanese.IsTriangulation.card_triangleSet`). `Japanese.triInradius` returns 0 for collinear points, but under `hconv` no triangle used is degenerate.

Lemma 32 (Poncelet)**.** *Under the hypotheses of Theorem 25, for $A \in \Gamma$:*

(a) A lies outside ω (`Poncelet3.radius_lt_dist_center_of_mem`*);*

(b) ω.`tangentsFrom` $A = \{AB, AC\}$ for the triangle of Theorem 25, and $AB \neq AC$ (`Poncelet3.tangentsFrom_eq_pair`*,* `Poncelet3.ncard_tangentsFrom`*);*

(c) for a tangent ℓ from A with nonzero direction vector v, Mathlib's Γ.`secondInter` A v is the unique point of $\Gamma \cap \ell$ other than A (`Poncelet3.secondInter_spec`*).*

By (b), the B and C of Theorem 25 are exactly the points produced by the classical construction, and by (c) that construction can be expressed with Mathlib's own `secondInter` (`Poncelet3.poncelet_construction_secondInter`).

9 Related work

Our searches covered the pinned Mathlib source, the current `docs/1000.yaml` of Mathlib's master branch (commit `6bd5e549`, 30 September 2026), GitHub code and repository search, Mathlib issues and pull requests, and web search. A search of the Lean Zulip chat on 30 September 2026 for "Carnot", "Viviani", "De Gua", "Japanese theorem" and "Poncelet" returned no relevant results. The searches are not exhaustive, and the claims below are "as far as we could determine". The per-project development notes record earlier rounds of these searches; where their early conclusions ("no earlier Lean proof found") predate the `lean-genius` findings below, this section supersedes them. External code is cited at the commit we read (see the bibliography).

Mathlib. In both the pinned version and current master, `docs/1000.yaml` lists Viviani's theorem, Carnot's theorem (inradius, circumradius), De Gua's theorem, both Japanese theorems and Poncelet's closure theorem with no declaration [10]. In the pinned `Mathlib/` source we found no occurrence of these names and no Cauchy–Binet formula, and the module list of current master has no Cauchy–Binet file. Mathlib does provide most of the vocabulary we use (Section 2), including `signedInfDist`, whose (1) does most of the work for Viviani and Carnot.

Euler's formula and Feuerbach's theorem. Wang's repository `wwylele/Feuerbach` [15] proves Feuerbach's theorem in Lean without `sorry`, and with it Euler's formula (`dist_incenter_circumcenter`) and Euler's inequality. His open Mathlib pull request #44236 [12] adds, among others, `dist_incenter_circumcenter_sq` and `dist_excenter_circumcenter_sq`. So Euler's formula, on which our Poncelet development depends, is not new in Lean; our copy comes from an independent development by the author. HOL Light's `100/feuerbach.ml` [8] proves Feuerbach's theorem in coordinates.

`lean-genius`. This public repository of Lean proofs [9] contains several related files. We read the five files cited below but did not build them; they contain no `sorry` and no declared `axiom`.

- `DeGuaTheorem.lean` proves the three-dimensional De Gua theorem for points of `Fin 3` $\to \mathbb{R}$, with cross-product areas, by one `linear_combination`. It is essentially our Theorem 8, so the three-dimensional case is not new in Lean.
- `VivianiTheorem.lean` proves Viviani's theorem for the unit equilateral triangle in coordinates, with explicit line equations. `VivianiTheoremOQ01OQ02.lean` proves that for any family of vectors whose sum is $(n+1)$ times a point g, the expression $\sum_i (a - \langle R^{-1}(g - v_i), P\rangle)$ equals $(n+1)a$; its link to the facets of a regular simplex is explained in comments but is not part of the statement. `VivianiTheoremOQ01OQ01.lean` proves a converse for triangles in $\mathbb{C}$.
- `CarnotTheorem.lean` proves the trigonometric identity $\cos A + \cos B + \cos C = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2}$ for reals with $A + B + C = \pi$. It does not mention a triangle, circumcentre or inradius. Several follow-up files (`CarnotTheoremOQ01*`) develop related trigonometric identities and inequalities; we only read their headers.

Our Viviani theorem uses Mathlib's affine simplices and facet distances in every positive dimension. Our De Gua theorem uses Gram-determinant volumes for indexed vertex families in arbitrary real inner product spaces, allowing degenerate configurations; it takes a point `O : E` and a family `P : Fin (m+1) → E`, not an `Affine.Simplex`.

Cauchy–Binet. The formula is not new as a formalisation, in Lean or elsewhere. In Lean: an open Mathlib pull request (#44209, labelled `awaiting-author` and `LLM-generated`) adds `Matrix.det_mul_eq_sum_det_submatrix` [11], and an earlier one (#40473) was closed without merging; `physlib` has a lemma `cauchyBinet` [13]; `rwst/Subspace-Theorems` has `det_mul_eq_sum_det_submatrix` [14]; and `kim-em/hex-dev` has, for its own Mathlib-free matrix library, the ordered-column expansion underlying Cauchy–Binet [7]: `det_columnSumMatrix_eq_sum_columnTuples` expands the determinant as a sum over all ordered column tuples, repeated-column summands are shown to vanish, and `det_gramMatrix_eq_sum_columnTuples` specialises this to a Gram matrix. That file's own header calls it the multilinearity form behind Cauchy–Binet; it does not state the sum over column subsets of products of two minors. We have not checked the axioms of these. In Isabelle/HOL, Divasón's AFP entry on the Smith normal form contains `Cauchy_Binet` [4]. In Coq, the CoqEAL library, built on MathComp, proves `BinetCauchy` in `theory/binetcauchy.v` [3]; it is described by Cano, Cohen, Dénès, Mörtberg and Siles [1]. Our proof is a self-contained alternative against the pinned Mathlib, nothing more.

Other systems. We searched the Isabelle AFP (through its GitHub mirror), HOL Light (a clone of the repository), GeoCoq, and the web, for each of the five theorems.

- *Isabelle AFP*: code search for "Viviani", "Carnot", "Poncelet", "Japanese", "trirectangular" and "Chapple" found nothing relevant. "Gua" is too common a substring to search this way.
- *HOL Light*: a case-insensitive search of the repository for "viviani", "poncelet", "carnot", "japanese", "de gua" and "cauchy-binet" found nothing.
- *GeoCoq*: code search for "Viviani", "Carnot", "Poncelet" and "Japanese" found nothing, and the single hit for "Gua" is unrelated.
- *Mizar*: we only checked the article file names of the current MML for "viviani", "carnot", "poncelet", "gua" and "binet", and found none; we did not search article contents.
- *Web*: search for formalisations of each theorem returned only informal sources.

Apart from reading the CoqEAL file cited above, we did not search the Coq/Rocq ecosystem beyond GeoCoq.

10 Verification

All five projects were built with prebuilt Mathlib on a separate build machine; the machine that prepared this paper did not run Lean. According to the development notes, Carnot, De Gua and Poncelet were rebuilt from an empty build directory (`rm -rf .lake/build; lake build`), and Viviani by a "clean `lake build`". The Japanese development was rebuilt from an empty build directory afterwards (log `clean_build.log`: 8714 jobs, 25 s, no warnings). Build logs are stored for Viviani, De Gua, Japanese and Poncelet; for Carnot only the notes record the build. None of this was reproduced on the machine that prepared this paper. Each build finished with no errors and no warnings. Wall-clock times: Carnot 15 s, Viviani 6 s, De Gua 10 s, Japanese 25 s (clean), Poncelet 30 s.

`#print axioms` was run on the headline theorems: 6 for Carnot, 8 for Viviani, 7 for De Gua, 29 for the Japanese development and 19 for Poncelet, 69 reports in all. The logs are kept in each repository (`ax.log`, `scratch/axioms.out` for Viviani, or `axioms.txt` for De Gua). In each case the driver is a scratch file outside the library (`scratch/`). Every report is `[propext, Classical.choice, Quot.sound]`. Every theorem in Table 1 either has its own report or occurs in the statement or proof of a theorem that has one. Where the development notes and the stored logs differ, the logs are authoritative. A grep over the library sources for `sorry`, `admit`, `native_decide`, `axiom`, `#eval`, `run_cmd`, `set_option`, `macro`, `elab`, `syntax`, `notation` and `import Lean` finds nothing. Each repository also has a `gate.sh` that rejects forbidden tokens, builds the library, generates its own `#print axioms` queries for every declaration cited in this paper (plus those queried during development) and fails on any other axiom. On the published commits it prints `PASS` with 12 (Carnot), 10 (Viviani), 10 (De Gua), 51 (Japanese) and 22 (Poncelet) declarations (at the version-2 commit, 28, 28, 23, 81 and 53; Section 10.1). `tests/tamper.sh` plants six kinds of defect in scratch copies (a `sorry`, a split `axiom`, a fake `#eval` report, a macro, a weakened statement, a wrapped extra axiom); in each repository the gate rejects all six, each with a nonzero exit status and the expected failure message.

Each project also has a numerical sanity check in `code/`, written in Python with the same definitions as the Lean code:

- *Carnot*: 20 000 random triangles, 14 514 of them obtuse, checking both the sum and $d_i = R\cos\alpha_i$; maximum relative error $4.5 \cdot 10^{-14}$.
- *Viviani*: 200 random simplices in each dimension 2–5 with arbitrary affine weights, and scaled regular simplices; maximum error $5.9 \cdot 10^{-12}$.
- *De Gua*: the three-dimensional and general forms in dimensions up to 7, and Cauchy–Binet including $k > n$ and $k = 0$; maximum relative error $2 \cdot 10^{-12}$.
- *Japanese*: all triangulations of 200 random convex cyclic n-gons for $n = 4, \ldots, 7$ (counts $2, 5, 14, 42$). The tree model and the maximal non-crossing sets are equal as families for $n \le 8$, and crossing agrees with interleaving for 293 088 chord pairs.
- *Poncelet*: 2 000 random configurations; the explicit points lie on Γ with all three distances equal to r (relative error $1.1 \cdot 10^{-12}$); with d perturbed, the third side misses ω.

The step that matters most, comparing the formal statements with the informal theorems, cannot be automated; Section 8 is meant to help with it. The author has checked each statement in Table 1 against its informal counterpart by hand.

10.1 Layers 14 and 15

Version 2 adds two further checks. Their files are `<Lib>/L14.lean` and `<Lib>/L15Classical.lean` in each repository (`Carnot`, `Viviani`, `DeGua`, `Japanese`, `Poncelet3`), at the new commit of each listed under Availability. Every theorem and lemma in them is listed in that repository's `gate.sh`, which prints `PASS` with 28 (Carnot), 28 (Viviani), 23 (De Gua), 81 (Japanese) and 53 (Poncelet) declarations, all with the three standard axioms only, and `tests/tamper.sh` still rejects all six planted defects. The earlier library files are byte-for-byte unchanged; only the root file of each library gains two import lines. The published `gate.sh` used GNU `mktemp` options and failed on macOS (BSD tools) before checking anything. The new commit uses a portable temporary file name, and the gate passes with BSD tools. The `wrap` test of `tests/tamper.sh` still needs GNU `sed`.

Layer 14: statements in Mathlib's language, with witnesses. In `<Lib>/L14.lean`, 18 of the numbered results are restated with Mathlib-only types and proved from the library theorems through definitional bridges: Theorems 2 and 3, with Mathlib's `signedInfDist` in place of `Carnot.signedSideDist` (`Carnot.l14_carnot`); Theorems 4 and 5, Proposition 6 and Corollary 7, with distances and heights written as `Metric.infDist` to the facet span, and the classical triangle form `Viviani.l14_viviani_triangle`; Theorems 8, 10 and 12, Lemma 11 and Corollary 13, with `triAreaSq` and `simplexVolSq` unfolded to the cross product and to det of a Gram matrix divided by $(k!)^2$; Lemma 16, Theorem 17 and Theorem 23, the last with `IsGeomTriangulation` written out (`Japanese.l14_japanese_polygon`); and Theorems 24, 25, 26 and 28, with `ConstructionCloses` unfolded into `tangentsFrom` and `IsTangent`. The elaborated type of each restatement contains no constant from our libraries. No restatement turned out false or needed an extra hypothesis. The Japanese restatements take an `AffineIndependent` argument, because a Mathlib triangle needs one; it follows from the orientation hypotheses and the conclusion does not depend on it. For every restated result with hypotheses, a witness theorem proves all hypotheses on an explicit configuration and proves that the conclusion is not degenerate: for Carnot, the obtuse triangle $0, 21, 6+8i$ in $\mathbb{C}$, whose three signed distances $3R/5$, $15R/17$, $-13R/85$ are distinct and of both signs (`Carnot.l14w_carnot`); for Viviani, a point with weights $(\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ in an equilateral triangle in $\mathbb{R}^3$, whose distances are positive and not all equal (`Viviani.l14w_viviani`); for De Gua, the corner $0, e_0, 2e_1, 3e_2$, with pairwise distinct positive squared face areas $\frac{49}{4}, 1, 9, \frac{9}{4}$, and for Cauchy–Binet a 2×3 by 3×2 product with determinant $7 \neq 0$; for the Japanese theorem, the quadrilateral $1, \frac{3+4i}{5}, -1, -i$ on the unit circle with its two distinct triangulations; for Euler's formula, the triangle $0, 4, 3i$, with $O \neq I$; and for the porism, its converse and the construction, circles with $d^2 = \frac{5}{4} = R(R-2r)$ and the explicit triangle $0, 4, 3i$ with its three points of tangency (`Poncelet3.l14w_poncelet`). Lemma 11 and Corollary 13 have no hypotheses and need no witness. Not restated: Theorem 19 and Propositions 21 and 22, which concern the tree model and the combinatorial definition (Theorem 23 covers the geometric statement); the correctness lemmas of Section 8; and the auxiliary lemma `Carnot.carnot_poly`. These were left out to stay within the time box.

Layer 15: comparison with the classical statements. Another agent collected textbook statements of the six topics from the literature, with citations (Johnson, Coxeter–Greitzer, Conant–Beyer, Fukagawa–Pedoe, Flatto and others). It cited them from memory and did not check page

numbers. An AI model given only these statements, with no access to our code or to this paper, wrote them in Lean with Mathlib notions and definitions of its own: 22 theorems and 3 well-formedness obligations, all with `sorry`. Its transcript and self-reported file list were audited (a transcript audit in the verification repository; the writer's run passed). A comparer then copied each statement verbatim and either proved it from the libraries, with no change to the library files, or classified it: 17 are *covered* (proved), 6 are covered only in a *narrower* sense, and 2 are *false as written* because the writer mis-stated them. None was classified as out of scope alone, but one of the two false statements (the incentre rectangle) is also outside the scope of this paper. The proofs are collected, with the writer's definitions and statements reproduced verbatim, in `<Lib>/L15Classical.lean`, namespace `<Lib>.Classical`, i.e. `Carnot.Classical` (and likewise `Viviani.Classical`, `DeGua.Classical`, `Japanese.Classical`, `Poncelet3.Classical`). All of them depend only on the three standard axioms.

New proofs (not in the main files):

- `Carnot.Classical.signedDistLine_face_eq` (also in Viviani): for any positive-dimensional simplex and any point in its affine span, the writer's signed distance to a facet equals `signedInfDist`.
- `Carnot.Classical.cov_TriSimplex1_b_acute`: for an acute triangle, Carnot's theorem with unsigned distances.
- `Carnot.Classical.cov_TriSimplex1_b_acute_inside`: the circumcentre of an acute triangle lies in its interior.
- `Viviani.Classical.height_two_eq`: the altitude of an equilateral triangle is $\frac{\sqrt{3}}{2}$ times the side; with it, Viviani with value $\frac{\sqrt{3}}{2}s$ (open, closed, signed in the plane).
- `DeGua.Classical.cov_TriSimplex2_a_coord_part`: the coordinate form of three-dimensional De Gua for `simplexVolSq`.
- `Japanese.Classical.quad_signs`, `Japanese.Classical.cov_TriSimplex4_b`: in the plane, strictly crossing diagonals together with positive orientation of ABC give the other three orientation hypotheses of Theorem 17, and the quadrilateral theorem holds for a convex quadrilateral (strictly crossing diagonals) with concyclic vertices, in any dimension.
- `Japanese.Classical.cov_TriSimplex4_a_obligation`: the triangles of a triangulation of a strictly convex polygon are non-degenerate.
- `Japanese.Classical.cov_TriSimplex4_b_obligation_part`: for pairwise distinct vertices, the four triangles of a convex quadrilateral are non-degenerate.
- `Poncelet3.Classical.cov_TriSimplex5_b`: the equality case of Euler's inequality, $R = 2r$ if and only if the triangle is equilateral.
- `Poncelet3.Classical.insphere_eq_of_isTangent`: three tangent side lines and centre inside Γ make ω the insphere of the triangle.
- `Poncelet3.Classical.cov_TriSimplex6_a`, `Poncelet3.Classical.cov_TriSimplex6_a_construction`, `Poncelet3.Classical.cov_TriSimplex6_b`, `Poncelet3.Classical.cov_TriSimplex6_b_converse`, `Poncelet3.Classical.cov_TriSimplex6_b_noclose`: the porism, the construction, the criterion and its two converses for triangles with vertices on Γ and insphere ω.

Narrower (proved part, and what is missing):

- De Gua in $\mathbb{R}^3$, its coordinate form, the n-simplex form and its coordinate form (four statements): proved for `simplexVolSq`; missing is the link to the writer's volume, Lebesgue measure of the hull in its span (and, for the n-simplex form, the indexing of facets by vertex sets).
- The Japanese theorem for polygons: proved for maximal non-crossing chord sets and orientation convexity (`Japanese.Classical.cov_TriSimplex4_a_part`); missing are the writer's dissections (cover with disjoint interiors) and side-line convexity.
- Two tangents from every point of Γ: proved only under $d^2 = R(R - 2r)$ (`Poncelet3.Classical.cov_TriSimplex6_a_obligation_part`); missing is the general case.

Writer statements false as written:

- Non-degeneracy of the four triangles of a convex quadrilateral: the writer's convex quadrilateral allows coincident vertices (counterexample $A = B = 0$, $C = D = 1$ in $\mathbb{R}$, `Japanese.Classical.not_classical_TriSimplex4_b_obligation`); true for distinct vertices, as above.
- The incentre rectangle: false for the same degenerate quadrilateral (`Japanese.Classical.not_classical_TriSimplex4_b_rectangle`); the corrected statement is not proved and is outside this paper.

The narrower cases are the gaps already stated in Sections 5 and 6 and in Section 11, and Lemma 32, which assumes $d^2 = R(R - 2r)$.

11 Limitations and further work

- *Duplication.* The Japanese development copies the Carnot files, and the Poncelet development copies the Euler files. A shared library, or upstreaming to Mathlib, would remove this. If PR #44236 is merged, the Euler copy should be replaced by Mathlib's lemmas.
- *Carnot with Mathlib's signed distance.* Theorem 2 is stated directly with `signedInfDist` only in the Layer-14 file (`Carnot.l14_carnot`), not in the main files.
- *Volumes.* `simplexVolSq` is a definition, not linked to Mathlib's Euclidean Hausdorff measure. For Viviani, we have not proved $h_i \operatorname{vol}(F_i) = n \operatorname{vol}(s)$, which would turn "equal heights" into "equal facet volumes". The Layer-15 comparison (Section 10.1) classifies the textbook De Gua statements, written with Lebesgue measure on the hull, as covered only in this narrower Gram-determinant sense.
- *Dissections.* The Japanese theorem is proved for maximal sets of chords with pairwise disjoint open segments. We have not shown that every dissection of the polygon into triangles with vertices among the p_l arises this way; that needs an area or measure argument. For this reason the Layer-15 comparison classifies the textbook polygon statement (triangulations as dissections, convexity by side lines) as covered only in the narrower sense (Section 10.1).
- *Convex position.* `hconv` is phrased with orientations, not with Mathlib's `Convex` or `convexHull`, and the polygon is indexed by $\mathbb{N}$ rather than `Fin n`. For planar quadrilaterals,

the Layer-15 lemma `Japanese.Classical.quad_signs` derives the other three orientation hypotheses from strictly crossing diagonals and positive orientation of ABC; the full quadrilateral proof chooses this orientation in the non-degenerate case and treats degenerate cases separately.

- *Poncelet.* Only triangles are treated. Poncelet's closure theorem for n-gons, even for $n = 4$ (Fuss's relation), would need a different argument or new certificates. The theorems of the main files assert side-line tangency only. The Layer-15 file (Section 10.1) proves, for $0 < r$ and $d < R - r$, the statements with an `Affine.Triangle` whose vertices lie on Γ and whose Mathlib insphere is ω; that the points of tangency lie on the side segments is not stated separately. That from every point of Γ there are exactly two tangents to ω, each meeting Γ again, is proved only under $d^2 = R(R - 2r)$ (Lemma 32).
- *Scope.* Lami's theorem, formalised separately by the author, is not part of this paper.

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